Heijmans: Project Context and Problem Understanding

1. Context

Heijmans needs reliable building information before starting a maintenance contract. This includes:

- Which assets are present.
- Their location within the building.
- Their condition.

At contract handover, the previous maintainer provides an asset inventory, often through Excel sheets and sometimes annotated floor plans. These records are frequently incomplete or outdated, requiring another building inspection.

This applied master's research project examines the potential of video-based 3D reconstruction and asset recognition to support inventory collection. Heijmans is the client; the people inspecting buildings and recording assets are the intended users.

2. Problem Description

At the start of a maintenance contract, Heijmans must establish a reliable inventory through building inspections and manual recording. This requires substantial time and effort. During the contract, changes are not consistently recorded, causing the inventory to become outdated and creating repeated work at the next handover.

The main issues are:

- Incomplete or outdated asset information.
- Time-consuming inspection and manual recording.
- Limited links between asset records and their locations.
- Difficulty keeping the inventory current.

The research problem is determining whether video-based 3D reconstruction and asset recognition can support reliable inventory collection with less manual effort. It remains unclear what model quality, capture completeness, processing speed and recognition accuracy can be achieved under relevant building conditions.

3. Project Objectives

Research objective: Evaluate how video-based 3D reconstruction and asset recognition can support building inventory collection for maintenance.

The research aims to:

- Compare selected methods in terms of model detail, completeness, processing time and asset recognition accuracy.
- Assess whether assets can be located by room and floor.

- Identify limitations and conditions affecting practical use.

Practical objective: Provide Heijmans with evidence to assess whether these methods can produce a reliable inventory with less collection and recording effort than the current process.

4. Research Questions

Main research question:

How can recent developments in world models and robotics algorithms support building maintenance through improved collection and updating of building asset information?

Subquestions:

1. Which concepts and methods from world models and robotics are relevant to collecting and updating this information?
2. How do these methods perform in terms of model quality, completeness, processing time and asset recognition accuracy?

The research will establish a working definition of *world models* and distinguish it from 3D reconstruction.

5. Expected Business and Technical Outcomes

Outcome	Expected contribution
Business value	Evidence for Heijmans to assess whether the investigated methods can reduce inventory effort while providing reliable asset information.
Technical insights	A comparison of selected methods, identifying their performance, limitations and conditions for use.
Research deliverables	A research report documenting the methods, experiments, results and recommendations.
Wider applicability	An assessment of which findings may apply to other building maintenance situations and which depend on the tested conditions.

6. Scope and Boundaries

The research focuses on building asset inventory collection for maintenance, using data captured by a person. The proposed scope is:

Within scope	Outside scope
Comparing selected methods for video-based 3D reconstruction and asset recognition.	Developing a production-ready maintenance system.
Evaluating model detail, completeness, processing time and recognition accuracy.	Autonomous robot navigation and data collection.
Investigating asset location by room and floor.	Centimetre-level positioning as a requirement.

The capture device could be a smartphone or Meta smart glasses. A study limited to one asset type is acceptable.

7. Potential Methods

Approach	Output
Photogrammetry	Camera poses, point clouds and surface meshes from overlapping images
Gaussian splatting	A scene represented by coloured 3D Gaussians for rendering different viewpoints
Semantic 3D mapping	Geometry associated with object categories and instances

8. Ethics and Data Management

- Agree on permission and conditions for recording at the selected building.
- Minimize capture of identifiable people and sensitive building information. Exclude or anonymize such content where necessary.
- Follow GDPR practises.

9. Planning

Sprint N	Duration
Sprint 1	Sept 23 - Oct 14
Sprint 2	Oct 15 - Nov 11
Sprint 3	Nov 12 - Dec 10
Sprint 4	Dec 11 - Jan 13
Sprint 5	Jan 14 - Jan 27

10. Stakeholders

Name	Email	Role and Responsibilities
Vladislav Stefanov	v.stefanov@student.fontys.nl	Executes and coordinates the project
Tom Langhorst	t.langhorst@fontys.nl	Fontys mentor
Ber Weeda	bweeda@heijmans.nl	Product owner
Dustin Bolte	dbolte@heijmans.nl	
Thomas Smits	tsmits2@heijmans.nl	